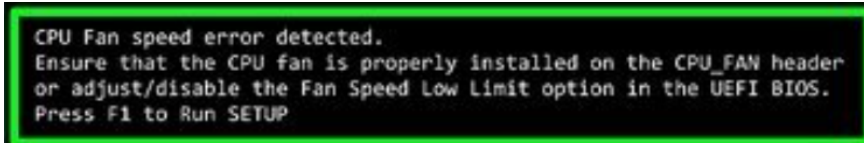


Assignment-3.3

“BIOS and POST troubleshooting”

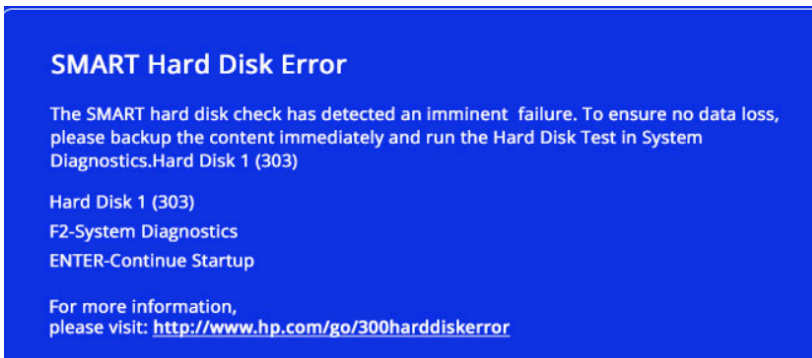
TASK 1 .List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

1.Message: CPU Fan Error or CPU Fan Speed Error



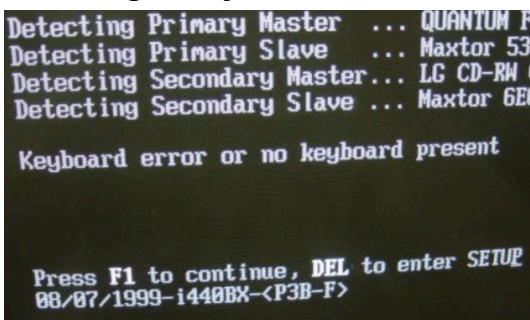
- **Hardware/Configuration Issue:** The BIOS detects that the main processor cooling fan is not spinning, is spinning too slowly, or is not plugged into the correct **CPU_FAN** header on the motherboard.

2.Message: SMART Hard Disk Error or Hard Disk S.M.A.R.T. Command Failed



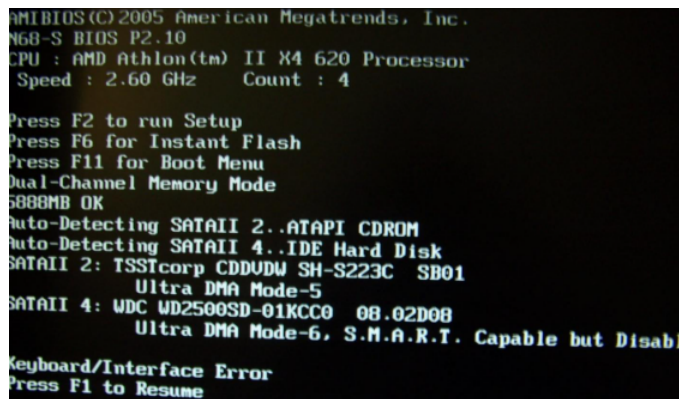
- **Hardware/Configuration Issue:** The storage drive's self-monitoring system detects an imminent mechanical or electrical failure on the hard drive or SSD.

3.Message: Keyboard Error or No Keyboard Present



- **Hardware/Configuration Issue:** The motherboard cannot detect a connected keyboard. This usually indicates an unplugged USB/PS2 cable, a broken USB port, or a failed keyboard controller circuit.

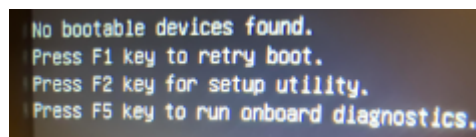
4.Message: Press F1 to Resume or Press F1 to Continue



AMIBIOS(C) 2005 American Megatrends, Inc.
M68-S BIOS P2.10
CPU : AMD Athlon(tm) II X4 620 Processor
Speed : 2.60 GHz Count : 4
Press F2 to run Setup
Press F6 for Instant Flash
Press F11 for Boot Menu
Dual-Channel Memory Mode
5888MB OK
Auto-Detecting SATAII 2..ATAPI CDROM
Auto-Detecting SATAII 4..IDE Hard Disk
SATAII 2: TSSTcorp CDDUDW SH-S223C SB01
Ultra DMA Mode-5
SATAII 4: WDC WD2500SD-01KCC0 08.02D08
Ultra DMA Mode-6, S.M.A.R.T. Capable but Disabled
Keyboard/Interface Error
Press F1 to Resume

- **Hardware/Configuration Issue:** A general POST warning triggered when a non-fatal hardware error occurs (such as a missing system clock setting, low battery voltage, or a missing cooling fan).

5.Message: No Boot Device Available or Reboot and Select Proper Boot Device



No bootable devices found.
Press F1 key to retry boot.
Press F2 key for setup utility.
Press F5 key to run onboard diagnostics.

- **Hardware/Configuration Issue:** The BIOS cannot locate a drive containing an operating system. This is caused by a disconnected hard drive/SSD, a corrupted OS bootloader, or an incorrect boot order selected in the BIOS.

TASK 2 Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

On an American Megatrends Inc. (AMI) BIOS, 3 short repeating beeps indicate a **Base 64KB RAM Failure**. This means the motherboard cannot read or write to the primary portion of the system memory during the initial boot check.

Two method of troubleshooting:

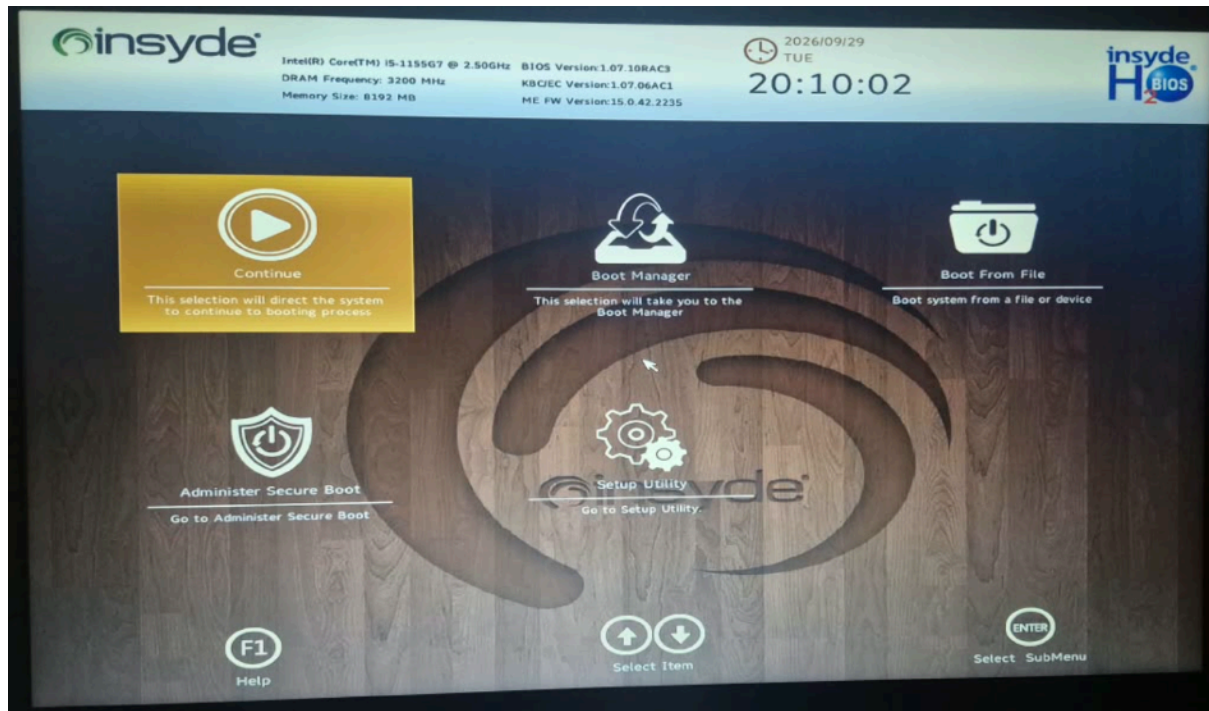
1. Reseat the RAM Modules:

Power off the PC, disconnect the power cord, open the case, and release the clips holding the RAM sticks. Pull the RAM sticks out, clear any dust from the slots, and firmly push them back into place until the clips click shut.

2. Test RAM Modules Individually (Single-Stick Testing):

If the PC has multiple RAM sticks, remove all of them except one. Try booting the PC with just that single stick inserted in the primary memory slot. If it boots, test the other stick to identify if one specific module or slot is defective.

TASK 3 Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



TASK 4 You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system. Think about the CMOS battery and BIOS settings reset.

A "CMOS checksum error" means the saved BIOS settings became corrupted or were erased because the CMOS chip lost power while the PC was turned off. The motherboard safely loaded its factory default settings so the PC could still turn on.

Cause: The small CR2032 coin-cell battery on the motherboard is dead and can no longer power the memory chip.

How to fix it:

1. Unplug the PC and replace the old CR2032 battery on the motherboard with a new one.
2. Turn on the PC, enter the BIOS menu (**Del** or **F2**), reconfigure the correct time and date, and press **F10** to save the settings and restart.